rich in energy. The relationship of the energy E of this photon to its frequency v is determined by Planck's formula, $E = hv$, where h is Planck's constant. The energy E is thus identical with the amount of the binding energy of the deuteron. This gamma radiation can actually be observed. Since we know how to measure the frequency v, it is actually possible to measure the binding energy of a deuteron. It amounts to 2.3 Mev. One might surmise that this energy is radiated in the form of several photons, rather than as a single one. However, it can be proved that such a process is actually far more improbable than the emission of one single photon.

There is, however, another, simpler method available for acquiring knowledge concerning the binding energy of atomic nuclei. According to Einstein's theory of relativity, there is a simple relationship between the mass of a body and its energy content. A relationship of this kind was known, in a special form, in pre-relativistic days, too; it follows from the electro-dynamics of moving objects. Hasenöhrl had already pointed out that the radiation enclosed in a cavity had a seemingly inert mass m, proportional to the energy of the system, m being proportional to E/c^2, where c is the velocity of light. But he failed to compute the proportionality factor correctly. The relationship between energy and mass was made clear by the theory of relativity, which—this is the decisive point—raised it to the status of a universal natural law, applicable not only to the theory of radiation, but to every other branch of physics as well. In other words, the following relation is true universally:

$$m = \frac{E}{c^2}$$

The meaning of this equation is: Any system having the energy content E has a mass m commensurate with this energy content, and the quantity of this mass is E/c^2. This conclusion has rather strange consequences. For instance, when a clock is being wound, it must become slightly heavier because energy is being stored in its spring. But the quantities of energy in this case are so small as to make the increase in the mass of the clock too minute to be capable of being demonstrated. The mass E/c^2 is much too small in comparison with the mass of the clock itself.

But this relationship between energy and mass can be put to practical use in nuclear physics, where the stored-up energy is of appreciable magnitude compared with nuclear masses. Expressed in the form $E = mc^2$, this relationship enables us to calculate the energy content, E, of a system from its mass, m. The velocity of light is known; it is almost exactly 300,000 km. per second, i.e. 3×10^{10} cm. per second. In atomic nuclei, the orders of magnitude are totally different from those met with in the above example of winding up a